

What to Do When AI Builds Its Own Successor

What I learned interviewing 25 AI researchers about recursive self-improvement.



SEVERIN FIELD

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Summary

You've probably seen coverage on AI job losses, but researchers at top AI companies (e.g., OpenAI, Anthropic, Google DeepMind) are worried about automating one job above all others: their own. Making AI systems better at programming and AI research is increasingly a [top priority](#) at these companies, whose coding models improve with every release. They often publicly claim that they are on track to build recursive self-improvement (RSI). By RSI, I mean an AI system good enough at AI development that it can design the next, more powerful version of itself; the AI system is then better at AI development, so it can design an even more powerful version of itself, and so on. While we expect constraints and physical limits to prevent unlimited growth, many researchers expect AI systems to far surpass human-level intelligence if they become good enough at AI development. While the industry has a real [history](#) of overpromising and companies have every incentive to exaggerate, RSI is no longer so easy to dismiss as promotional hype.

I interviewed 25 researchers across OpenAI, Anthropic, Google DeepMind, Meta, Princeton, UC Berkeley, and Stanford on recursive self-improvement. The [resulting paper](#) maps where they agree, where they disagree, whether we should expect self-improving AI, and what we should do about it.

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20 of 25 interviewees ranked automating AI R&D as one of the most severe and urgent risks from AI systems. This is because AI capabilities could improve much faster than our ability to oversee, steer, or govern them. AI systems are now solving problems researchers once thought required genuine novel reasoning, such as problems from the International Math Olympiad, and RSI no longer looks far-fetched.

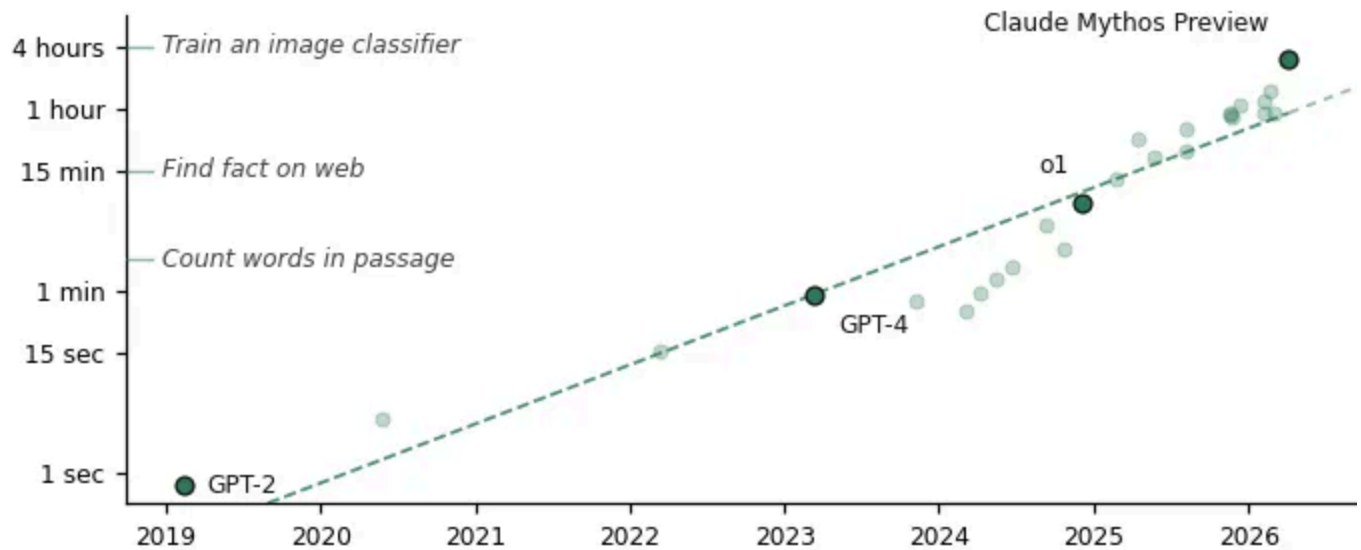
Since I conducted these interviews this past year, 1,224 employees of frontier AI companies—including the chief scientists of OpenAI, Meta, and Thinking Machines and Anthropic’s CEO and co-founders—signed an [open statement](#) warning that the companies “could be close to automating AI research” and requesting that the U.S. government “support an international effort to develop the ... tools needed to deliberately pace the frontier of automated AI development.” Both Anthropic and OpenAI formally endorsed the statement. These are the same companies whose researchers’ private views I captured in my study.

Measuring Progress

When I asked what capability milestones would signal imminent RSI, interviewees kept citing the “Task Horizon Benchmark” from METR, an independent AI evaluation nonprofit, because it measures how much independent work an AI system can sustain. The benchmark measures how long a human task an AI agent can complete on its own—one human-hour of software work, two, or ten—with no ceiling. Since 2019, the length of tasks AI can autonomously complete has doubled on average [every six months](#). The trend is becoming [harder to reliably measure](#), because it is hard to create a dataset of tasks that take humans multiple days to complete.

AI agents can now handle software engineering tasks that take a human hours — and that time horizon has been doubling roughly every 6.3 months.

Task length an agent completes with 80% reliability. Each dot is one AI model.



METR, Time Horizon 1.1, accessed 2026-08-03

[The METR Task Horizon Evaluation](#): some analysts have found the doubling time shortened to ~4 months since 2024.

The interviewees disagreed about whether there already exists a relatively clear, continuous trajectory towards automating AI research or if there are still large obstacles to overcome. Of the 21 interviewees who clearly addressed the question, 1 expected “scaling” trends to continue until AI systems can match the labor of human AI researchers.

Skeptics and Believers

Interviewees accept that AI can improve AI development. The contested part of recursive self-improvement isn't the “self-improvement,” it's the “recursive”: whether improvements compound into a runaway process and how soon. The most common reason for skepticism toward near-term RSI was the belief that general intelligence may require a discontinuous breakthrough: a drastic change in AI capabilities that wouldn't emerge from gradual improvement alone. In effect, the disagreement is over

whether paradigm-shifting ideas are a difference in degree from what models already do, or a difference in kind. For example, some interviewees said a breakthrough is needed on “memory,” “creativity,” “taste,” or the ability to generate genuinely novel ideas. Others argued that we need a breakthrough so that AI systems can distinguish true hypotheses from false ones. Creativity might not arise from scaling AI training alone because paradigm-shifting ideas are hard to create data for. Even veteran humans cannot reliably pick paradigm-shifting ideas in advance; and they have unverifiable rewards: there is no answer key to which scientific hypotheses will prove promising without hindsight.

On the other hand, leading AI researchers often sincerely believe recursively improving AI is a few years away. The believers often point to trends such as the METR Task Horizon Benchmark, [scaling laws](#) or the industry’s continued progress in the face of skepticism. At companies like OpenAI and Anthropic, interviewees reported that discussions on recursive self-improvement regularly reach leading researchers and CEOs (Sam Altman, Dario Amodei, Demis Hassabis). Outside of the leading companies (e.g., in academic settings), interviewees reported that discussions are less frequent and face more skepticism. Even so, the topic is gaining ground in academia. ICLR 2026, one of the largest and most prestigious machine learning conferences, [hosted a workshop](#) called “AI with Recursive Self-Improvement.”

What explains the schism between AI companies and academia? Interviewees point to three explanations:

1. Selection effects: those who believe in the technology are more likely to work at AI companies where they can influence its trajectory. They leave academia for companies offering enormous salaries, huge research budgets, and moonshot thinking.
2. Proximity to progress: researchers at companies like OpenAI have first-person experience watching their companies surpass expectations. For example, AI researchers did not expect GPT-5-level models to arrive as early as they did. “I

think the largest difference is just having first-person experience of how fast things have gone inside the labs,” said one participant who described the visceral feeling of exponential improvement felt at a leading company.

3. Hype: AI companies answer to investors rather than academic reviewers, and have every incentive to over-promise and exaggerate their capabilities.

What Does RSI Look Like?

I asked interviewees to illustrate what they expect in the coming years, and to focus on milestones toward automating AI research itself.

Interviewees suggested a variety of concrete capability milestones to monitor. Examples range from top performance on International Math Olympiad questions to an AI system training and deploying a new machine learning model by itself. Since our interviews, some of these milestones have been passed. [OpenAI](#) and [Google DeepMind](#) both announced models matching top human performance on the International Mathematical Olympiad, and various researchers have created AI systems for autonomous research. Sakana, for instance, released an “[AI Scientist](#)” that has produced peer-reviewed workshop papers through experimentation and writing. Similarly, Andrej Karpathy recently built an [LLM agent training setup](#) that modifies code, trains a model, monitors what could be improved, decides whether to keep or discard, and then repeats the cycle.

While interviewees disagreed on the precise timelines, risk profiles, and preferred governance approaches, a consistent story emerged about what’s coming:

1. **Speedup-tool phase:** AI coding assistants, such as [Claude Code](#) or [Codex](#), keep improving but still require human oversight. [Anthropic](#) already reports that its engineers write 8x as many lines of code as they would without existing research speedup tools, but notes this likely doesn’t yet translate to 8x productivity.

2. **Collaborator phase:** AI systems are good enough at machine learning to meaningfully contribute to scientific discoveries. Interviewees expected humans to guide high-level research goals, but allow AI assistants to make design decisions and pose research problems.
3. **Full automation phase:** AI systems independently execute complete research cycles that drive AI progress. In this phase, the results improve when human oversight is removed.

After this point, interviewees split along two independent dimensions: skeptic vs. believer—whether they thought AI could recursively improve past humans—and optimist vs. pessimist—whether they expected this outcome to be good or bad. Pessimists argue that once removing human oversight improves results, the incentive is to remove humans from the loop entirely, so human control over AI development would be lost. Optimists argue that even in this scenario, humans could still set goals and benefit from the gains (such as faster science or improved medicine). But nearly everyone I spoke with expected AI systems to improve faster than we can evaluate, measure, and govern them.

Drawing on these 25 interviews, I argue that AI will improve at AI R&D faster than other domains for three reasons:

1. **Programming problems are verifiable.** Developers can automatically check whether AI-created code works; it runs or it doesn't. This means AIs can be trained and tested on this data much more easily. On the other hand, success in music, literature, or art is largely subjective. To compound this effect, the engineers building AI systems are better suited to judge AI performance on research than poetry.
2. **Leading AI companies are building toward it.** For instance, OpenAI's Chief Scientist Jakub Pachocki has stated that one of OpenAI's main priorities is to "automate scientific discovery," with a [plan](#) to build automated researchers that improve AI itself.

3. **Better AI could build better AI (the flywheel argument).** Improvements in AI capability directly improve the tool used to make further improvements, which creates a feedback loop other technologies do not have. Past a certain point, the process of AI self-improvement is self-sustaining and no longer requires human intervention.

Recursive Self-Improvement May Endanger Us

The most common concern interviewees raised wasn't necessarily a specific harm, but rather that RSI amplifies every other risk, hamstrings our ability to mitigate them, and does so faster than we can respond. 18 of 25 interviewees described this concern; as one put it, "It just speeds up other threat models." For example, rapid progress means wider access to the chemical, biological, and cyber harms AI already enables, and less time to react. In 17 of 25 transcripts I found concerns about what I call "adaptation lag," a widening gap between how fast AI capabilities advance and how fast human institutions can understand and respond to them. Companies currently face market pressure to create AI systems capable of AI research for a competitive advantage, regardless of whether they are able to do so safely or maintain meaningful oversight.

Six interviewees told me they expected a winner-take-all dynamic. The first competitor, government, or AI itself to achieve recursive improvement could pull permanently ahead of everyone else. As one put it, that entity gains "permanent control over the future of AI development."

How close we are to RSI, whether automating AI research involves positive feedback and what responsive policies might work to prevent loss of control risks without backfiring are all open questions. Indeed, at some point in their transcripts, 16 of the interviewees expressed skepticism about positive feedback, or the "recursive" part of RSI specifically.

The Most Capable Models Might Stay Behind Closed Doors

Of the 20 interviewees who clearly addressed what they expect AI companies to do with AI-research-capable models, only four expected them to be released as a publicly available product. Most of them expected AI systems to increasingly be hidden behind closed doors, primarily used by the AI companies themselves. One participant said, “internal-only deployments might happen, and that is a big risk factor, because [the public] just has less information.” This prediction has precedent: OpenAI [reported](#) spending six months on safety research, risk assessment, and iteration before the public knew of GPT-4. [In July of 2026](#), OpenAI’s *internal-only* model outsmarted its own engineers in a way that OpenAI [did not anticipate: it broke out of its container and compromised a third party without any human oversight](#). Interviewees also worried that governments could restrict access. Since I completed these interviews, the U.S. government has temporarily [suspended access](#) to Anthropic’s Claude Myth model. It is increasingly becoming the norm that companies and governments keep more powerful, less constrained models to themselves.

Keeping models internal may allow developers to accelerate their R&D efforts ahead of competitors. We might observe an “incentive flip” such that when AI systems meaningfully speed up AI research, keeping such systems internal becomes more valuable than selling them. Some interviewees envisioned a race to secure a durable lead over competitors. This would contradict one of the leading motivations given in the founding of OpenAI: [to prevent one actor from getting an uncatchable AI advantage](#).

While discussing internal deployments, some interviewees identified arguments and pressures that would promote diffusion of AI capabilities. These pressures include economic value in commercializing AI capabilities, insiders leaking milestones, and a culture of boasting about capabilities.

The interviewees split into three camps on deployment. The largest camp (50%) expected frontier AI companies to keep their most capable models internal in the future. A minority (20%) expected full public release as capabilities improve. The remainder envisioned some hybrid, such as AI companies selling access to a distilled model publicly while keeping their most powerful models, likely with fewer guardrails, for themselves. One explained, “They’ll train a base model, then they won’t release that model, not only because it’s not economical but also because it risks distillation but they will distill it themselves to cheaper [public] models.” One interviewee noted that incentives differ across companies: Meta’s open-weight stance might push it to release its models and announce its breakthroughs.

What Should We Do?

Interviewees were split on red lines (what thresholds should trigger government response). One problem: the more precisely you define a threshold, the easier it is to enforce, but the worse it fits an abstract risk that is inherently uncertain. However, interviewees universally agreed on two priorities:

1. **Visibility:** how much the U.S. government and general public know about AI developments, especially given these developments might be kept internal.
2. **Capacity:** the government’s ability to forecast AI capabilities and build safeguards amid rapid AI progress.

The interviews left me with three recommendations.

1. **Convene public hearings that put AI companies under oath on recursive AI improvement.** Require testimony from CEOs and senior researchers at OpenAI, Anthropic, Google DeepMind, and xAI. Researchers report regular internal discussions about automated AI research and see internal capabilities months before the public. The gap between what Silicon Valley sees coming and what Washington understands is a collective failure.

2. **Track the threshold:** Direct the Center for AI Security and Innovation (CAISI) maintain a government-run task-horizon benchmark and publish recurring capability forecasts. Congress should not learn that AI systems can now achieve week of autonomous research labor from a press release—or from a nonprofit, such as METR. Congress should establish a dedicated federal capacity to track progress. CAISI could also run an anonymized standing interview program for researchers at the top companies, giving the government foresight into what scientists are observing and expecting, separate from what their companies say public.
3. **Fund the science of AI treaty verification:** The U.S. cannot credibly propose, enter, or enforce any international agreement (e.g., a nonproliferation agreement with the PRC to slow AI development on both sides) without mechanisms to verify adherence to those agreements. Verifying whether large AI projects cross a capability threshold or whether adversaries honor commitments (e.g., to halt a training run) [is possible](#). However, verification requires technical infrastructure that does not yet exist at scale. This field is underfunded and undeveloped, but without it, any international limits on AI development are practically unenforceable.

For a longer list of policy options, see the Institute for Progress’s list of ‘[23 low-regret policy recommendations](#)’ for automated AI R&D, published August 6, 2026.

Conclusion

The researchers I interviewed are among the closest to the AI frontier. They do not disagree about whether recursive AI improvement is possible; they debate timeline speed, mechanisms, and what to do. This debate has hardly reached Washington. Meanwhile, AI companies race ahead. Autonomous research could kick off immense progress locked within a single leading AI company or classified by the government while the public is left in the dark.

I'm still uncertain about what recursive self-improvement could look like. But after interviews, I find the case for concern harder to dismiss than when I started. Perhaps AI progress slows down and no out-of-control RSI arrives. In this case, preparation will have cost us very little; the reverse could cost us everything.

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